1. Setup e Imports

```
import json as _json
import torch
import torch.nn as nn
import torch.nn.functional as F
from torch.utils.data import Dataset, DataLoader

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from tqdm import tqdm
import os
import sys
from pathlib import Path

# Agregar path del proyecto para imports
sys.path.append(os.path.abspath('../../..'))
from sae.models.sae import SparseAutoencoder
from sae.tools.naming_utils import save_checkpoint, get_model_name, get_extras_id, get_checkp
from sae.tools.experiment_utils import get_experiment_dir, get_plots_dir, get_metrics_dir, ge

# Configuración de visualización para Quarto/PDF
pd.set_option('display.width', 100)
pd.set_option('display.max_columns', None)
np.set_printoptions(linewidth=100, precision=4)
os.environ['COLUMNS'] = '100'

# Configuración de reproducibilidad
torch.manual_seed(42)
np.random.seed(42)

# Device
device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
```

```
# Print config con formato profesional
config_df = pd.DataFrame({
    'Configuración': ['Device', 'Seed', 'Output Width'],
    'Valor': [str(device), '42', '100']
})

print('\n' + '='*50)
print(' CONFIGURACIÓN INICIAL')
print('='*50)
print(config_df.to_string(index=False))
print('='*50)
```

```
==================================================
 CONFIGURACIÓN INICIAL
==================================================
Configuración Valor
       Device  cuda
         Seed    42
 Output Width   100
==================================================
```

2 Metadata para Naming de Checkpoints

```
# Metadata para naming de checkpoints (separada de hiperparámetros)
NAMING_META = {
    'variante': 'standard',          # Arquitectura del SAE
    'tecnica': 'l1-decay',                 # L1 con penalty annealing
    'capa': 6,                       # Layer del modelo OthelloGPT
    'juegos': 500,                  # Número de partidas
    'base_path': 'C:\\Users\\Esposa\\Documents\\Repos\\sae-othello-gpt',  # Path base para cl
    'experiment_base_path': os.path.abspath('../../../experiments') ,      # sae/experiments/
    'extras': {
        'expansion': 32,
        'sparsity_coef': 5e-4,
        'epochs': 200,
        'patience': 15,
        'learning_rate': 1e-3,
        'sparsity_decay': 0.95,
```

```python
        'batch_size': 256
    }
}

print('\n Metadata de naming:')
for key, value in NAMING_META.items():
    print(f'  {key}: {value}')
```

```
 Metadata de naming:
  variante: standard
  tecnica: l1-decay
  capa: 6
  juegos: 500
  base_path: C:\Users\Esposa\Documents\Repos\sae-othello-gpt
  experiment_base_path: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments
  extras: {'expansion': 32, 'sparsity_coef': 0.0005, 'epochs': 200, 'patience': 15, 'learning
```

```python
extras_id = get_extras_id(
    get_checkpoint_dir(
        NAMING_META['base_path'],
        NAMING_META['variante'],
        NAMING_META['tecnica'],
        NAMING_META['capa'],
        NAMING_META['juegos']
    ),
    NAMING_META['extras']
) if NAMING_META.get('extras') else None

print(f'extras_id: {extras_id}')
```

```
extras_id: ex5
```

3. Configuración del SAE

Las activaciones ya fueron extraídas previamente con `sae/activations/run_extraction.py`. Aquí especificamos qué archivo cargar y los hiperparámetros del SAE.

```python
# Configuración del SAE
CONFIG = {
    # Arquitectura
    'input_dim': 512,
    'hidden_dim': 16384,
    'tied_weights': False,


    # Entrenamiento
    'batch_size': 256,
    'num_epochs': 200,
    'learning_rate': 1e-3,
    'sparsity_coef': 5e-4,
    'sparsity_decay': 0.95,

    # Early Stopping
    'early_stopping': True,
    'patience': 15,

    # Datos
    'activations_file': Path(f"../../../activations/data/layer{NAMING_META['capa']}_{NAMING_M
    'save_dir': Path('./saved_model'),
    'plots_dir': get_plots_dir(
        NAMING_META['experiment_base_path'],
        NAMING_META['variante'],
        NAMING_META['tecnica'],
        NAMING_META['capa'],
        NAMING_META['juegos'],
        extras_id=extras_id
    )
}

# Crear directorio local del modelo (plots_dir ya fue creado por get_plots_dir)
CONFIG['save_dir'].mkdir(parents=True, exist_ok=True)

# Mostrar configuración con formato profesional
config_data = {
    'Categoría': [],
    'Parámetro': [],
    'Valor': []
}
```

```python
# Arquitectura
for key in ['input_dim', 'hidden_dim', 'tied_weights']:
    config_data['Categoría'].append('Arquitectura')
    config_data['Parámetro'].append(key)
    config_data['Valor'].append(str(CONFIG[key]))

# Entrenamiento
for key in ['batch_size', 'num_epochs', 'learning_rate', 'sparsity_coef', 'sparsity_decay']:
    config_data['Categoría'].append('Entrenamiento')
    config_data['Parámetro'].append(key)
    config_data['Valor'].append(str(CONFIG[key]))

# Early Stopping
for key in ['early_stopping', 'patience']:
    config_data['Categoría'].append('Early Stopping')
    config_data['Parámetro'].append(key)
    config_data['Valor'].append(str(CONFIG[key]))

config_table = pd.DataFrame(config_data)

print('\n' + '='*70)
print(' CONFIGURACIÓN DEL SAE')
print('='*70)
print(config_table.to_string(index=False))
print('='*70)

print(f'\nArchivo de activaciones: {CONFIG["activations_file"]}')
print(f'Expansión: {CONFIG["hidden_dim"]/CONFIG["input_dim"]:.1f}x')
print(f'Plots dir: {CONFIG["plots_dir"]}')
print('='*70)
```

```
======================================================================
 CONFIGURACIÓN DEL SAE
======================================================================
     Categoría      Parámetro  Valor
  Arquitectura      input_dim    512
  Arquitectura     hidden_dim  16384
  Arquitectura   tied_weights  False
 Entrenamiento     batch_size    512
 Entrenamiento     num_epochs    200
 Entrenamiento  learning_rate  0.001
```

```
 Entrenamiento  sparsity_coef 0.0005
 Entrenamiento sparsity_decay   0.95
Early Stopping early_stopping   True
Early Stopping       patience     15
======================================================================

Archivo de activaciones: ..\..\..\activations\data\layer6_500games.npy
Expansión: 32.0x
Plots dir: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standa
decay_standard\500games\ex5\plots
======================================================================
```

4. Dataset de Activaciones (Train/Validation Split 80-20%)

```python
class ActivationsDataset(Dataset):
    """Dataset para cargar activaciones extraídas"""

    def __init__(self, activations_path):
        self.activations = np.load(activations_path)

    def __len__(self):
        return len(self.activations)

    def __getitem__(self, idx):
        return torch.tensor(self.activations[idx], dtype=torch.float32)

# Cargar dataset completo
full_dataset = ActivationsDataset(CONFIG['activations_file'])

# Información del dataset con formato profesional
dataset_info = pd.DataFrame({
    'Propiedad': ['Total muestras', 'Shape', 'Dtype', 'Tamaño en memoria'],
    'Valor': [
        f"{len(full_dataset):,}",
        str(full_dataset.activations.shape),
        str(full_dataset.activations.dtype),
        f"{full_dataset.activations.nbytes / 1024**2:.2f} MB"
    ]
})
print('\n' + '='*50)
print(' DATASET DE ACTIVACIONES')
```

```
print('='*50)
print(dataset_info.to_string(index=False))
print('='*50)

# Split train/validation (80-20)
train_size = int(0.8 * len(full_dataset))
val_size = len(full_dataset) - train_size
train_dataset, val_dataset = torch.utils.data.random_split(
    full_dataset,
    [train_size, val_size],
    generator=torch.Generator().manual_seed(42)
)

# Resumen del split con formato profesional
split_info = pd.DataFrame({
    'Conjunto': ['Train', 'Validation', 'Total'],
    'Muestras': [
        f"{len(train_dataset):,}",
        f"{len(val_dataset):,}",
        f"{len(full_dataset):,}"
    ],
    'Porcentaje': [
        f"{len(train_dataset)/len(full_dataset):.1%}",
        f"{len(val_dataset)/len(full_dataset):.1%}",
        "100.0%"
    ]
})
print('\n' + '='*50)
print(' SPLIT DE DATOS (80-20)')
print('='*50)
print(split_info.to_string(index=False))
print('='*50)

# Crear DataLoaders
train_loader = DataLoader(
    train_dataset,
    batch_size=CONFIG['batch_size'],
    shuffle=True,
    num_workers=0,
    pin_memory=(device.type == 'cuda')
)
```

```python
val_loader = DataLoader(
    val_dataset,
    batch_size=CONFIG['batch_size'],
    shuffle=False,  # No shuffle para validación
    num_workers=0,
    pin_memory=(device.type == 'cuda')
)

# Resumen de DataLoaders
loader_info = pd.DataFrame({
    'DataLoader': ['Train', 'Validation'],
    'Batches': [len(train_loader), len(val_loader)],
    'Batch Size': [CONFIG['batch_size'], CONFIG['batch_size']],
    'Shuffle': ['Sí', 'No']
})

print('\n' + '='*50)
print(' DATALOADERS')
print('='*50)
print(loader_info.to_string(index=False))
print('='*50)
```

```
==================================================
 DATASET DE ACTIVACIONES
==================================================
        Propiedad        Valor
   Total muestras       29,500
            Shape (29500, 512)
            Dtype      float32
Tamaño en memoria     57.62 MB
==================================================

==================================================
 SPLIT DE DATOS (80-20)
==================================================
  Conjunto Muestras Porcentaje
     Train   23,600      80.0%
Validation    5,900      20.0%
     Total   29,500     100.0%
==================================================
```

```
==================================================
 DATALOADERS
==================================================
DataLoader  Batches  Batch Size Shuffle
     Train       47         512      Sí
Validation       12         512      No
==================================================
```

5. Modelo: Sparse Autoencoder

Modelo importado desde `sae.models.sae.SparseAutoencoder`

Arquitectura:

- **Encoder**: Linear → ReLU (sparsity natural)- **Decoder**: Linear (reconstrucción)

```
# Crear modelo desde el módulo
model = SparseAutoencoder(
    input_dim=CONFIG['input_dim'],
    hidden_dim=CONFIG['hidden_dim'],
    tied_weights=CONFIG['tied_weights']
).to(device)

# Calcular parámetros
num_params = sum(p.numel() for p in model.parameters())
encoder_params = sum(p.numel() for p in model.encoder.parameters())
if CONFIG['tied_weights']:
    decoder_params = CONFIG['input_dim']  # Solo bias
else:
    decoder_params = sum(p.numel() for p in model.decoder.parameters())

# Resumen del modelo con formato profesional
model_info = pd.DataFrame({
    'Componente': ['Input', 'Encoder', 'Hidden (Latent)', 'Decoder', 'Output', 'Total'],
    'Dimensión': [
        CONFIG['input_dim'],
        f"{CONFIG['input_dim']} → {CONFIG['hidden_dim']}",
        CONFIG['hidden_dim'],
        f"{CONFIG['hidden_dim']} → {CONFIG['input_dim']}",
        CONFIG['input_dim'],
        '-'
    ],
    'Parámetros': [
```

```python
        '-',
        f"{encoder_params:,}",
        '-',
        f"{decoder_params:,}",
        '-',
        f"{num_params:,}"
    ]
})

print('\n' + '='*70)
print(' ARQUITECTURA DEL MODELO')
print('='*70)
print(model_info.to_string(index=False))
print('='*70)
print(f'\n Factor de expansión: {CONFIG["hidden_dim"]/CONFIG["input_dim"]:.1f}x')
print(f' Tied weights: {"Sí" if CONFIG["tied_weights"] else "No"}')
print(f' Device: {device}')
print('='*70)
```

```
======================================================================
 ARQUITECTURA DEL MODELO
======================================================================
     Componente   Dimensión Parámetros
          Input         512          -
        Encoder 512 → 16384  8,404,992
Hidden (Latent)       16384          -
        Decoder 16384 → 512  8,389,120
         Output         512          -
          Total           - 16,794,112
======================================================================

 Factor de expansión: 32.0x
 Tied weights: No
 Device: cuda
======================================================================
```

6. Función de Pérdida

$$\textbf{Loss} = \textbf{MSE(reconstrucción)} + \quad \times \textbf{L1(activaciones)}$$

- **MSE**: Mide qué tan bien reconstruimos las activaciones originales
- **L1**: Penaliza activaciones grandes → fuerza sparsity

```
def loss_function(x, reconstruction, hidden, sparsity_coef):
    """
    Pérdida compuesta: MSE + L1 penalty

    Args:
        x: Activaciones originales
        reconstruction: Activaciones reconstruidas
        hidden: Activaciones latentes (sparse)
        sparsity_coef: Peso de la penalización L1
    """
    # Error de reconstrucción
    mse_loss = F.mse_loss(reconstruction, x)

    # Sumar sobre las features (dim=1), promediar sobre el batch
    sparsity_loss = hidden.abs().sum(dim=1).mean()

    # Pérdida total
    total_loss = mse_loss + sparsity_coef * sparsity_loss

    return total_loss, mse_loss, sparsity_loss

# Optimizador
optimizer = torch.optim.Adam(model.parameters(), lr=CONFIG['learning_rate'])

# Resumen del optimizador
optim_info = pd.DataFrame({
    'Parámetro': ['Optimizador', 'Learning Rate', 'Sparsity Coef (inicial)', 'Sparsity Decay
    'Valor': [
        'Adam',
        f"{CONFIG['learning_rate']:.0e}",
        f"{CONFIG['sparsity_coef']:.0e}",
        f"{CONFIG['sparsity_decay']:.2f}"
    ]
})

print('\n' + '='*50)
print(' OPTIMIZACIÓN')
print('='*50)
print(optim_info.to_string(index=False))
print('='*50)
```

```
==================================================
 OPTIMIZACIÓN
==================================================
             Parámetro Valor
           Optimizador  Adam
         Learning Rate 1e-03
Sparsity Coef (inicial) 5e-04
        Sparsity Decay  0.95
==================================================
```

7. Loop de Entrenamiento

```python
# Historial de entrenamiento
history = {
    'train_total_loss': [],
    'train_mse_loss': [],
    'train_sparsity_loss': [],
    'train_l0_sparsity': [],
    'train_fraction_active': [],
    'val_total_loss': [],
    'val_mse_loss': [],
    'val_sparsity_loss': [],
    'val_l0_sparsity': [],
    'val_fraction_active': [],
    'sparsity_coef_history': []  # Guardar evolución del coeficiente L1
}

# Early stopping variables
best_val_mse = float('inf')
best_epoch = 0
epochs_no_improve = 0

print(f'\n Iniciando entrenamiento: {CONFIG["num_epochs"]} épocas máximo')
print(f' L1 decay: {CONFIG["sparsity_coef"]:.2e} × {CONFIG["sparsity_decay"]:.2f}^época')
print(f'   Early stopping: patience={CONFIG["patience"]} épocas')
print('=' * 70)

for epoch in range(CONFIG['num_epochs']):
```

```
# ========== L1 DECAY ==========
# Reducir sparsity_coef exponencialmente
current_sparsity_coef = CONFIG['sparsity_coef'] * (CONFIG['sparsity_decay'] ** epoch)
history['sparsity_coef_history'].append(current_sparsity_coef)

# ========== ENTRENAMIENTO ==========
model.train()

epoch_total_loss = 0
epoch_mse_loss = 0
epoch_sparsity_loss = 0
epoch_l0 = 0
epoch_fraction_active = 0

# Barra de progreso
pbar = tqdm(train_loader, desc=f'Época {epoch+1}/{CONFIG["num_epochs"]} [Train] L1={curre

for batch in pbar:
    # Mover al device
    x = batch.to(device)

    # Forward pass
    reconstruction, hidden = model(x)

    # Calcular pérdida con sparsity_coef actualizado
    total_loss, mse_loss, sparsity_loss = loss_function(
        x, reconstruction, hidden, current_sparsity_coef
    )

    # Backward pass
    optimizer.zero_grad()
    total_loss.backward()

    optimizer.step()

    # Métricas
    with torch.no_grad():
        l0 = (hidden > 0).float().sum(dim=1).mean().item()
        fraction_active = (hidden > 0).float().mean().item()

    # Acumular
    epoch_total_loss += total_loss.item()
```

```
        epoch_mse_loss += mse_loss.item()
        epoch_sparsity_loss += sparsity_loss.item()
        epoch_l0 += l0
        epoch_fraction_active += fraction_active

        # Actualizar barra
        pbar.set_postfix({
            'loss': f'{total_loss.item():.4f}',
            'mse': f'{mse_loss.item():.4f}',
            'l0': f'{l0:.1f}'
        })

    # Promedios de entrenamiento
    num_train_batches = len(train_loader)
    avg_train_total_loss = epoch_total_loss / num_train_batches
    avg_train_mse_loss = epoch_mse_loss / num_train_batches
    avg_train_sparsity_loss = epoch_sparsity_loss / num_train_batches
    avg_train_l0 = epoch_l0 / num_train_batches
    avg_train_fraction_active = epoch_fraction_active / num_train_batches

    # ========== VALIDACIÓN ==========
    model.eval()

    val_total_loss = 0
    val_mse_loss = 0
    val_sparsity_loss = 0
    val_l0 = 0
    val_fraction_active = 0

    with torch.no_grad():
        for batch in val_loader:
            x = batch.to(device)

            # Forward pass
            reconstruction, hidden = model(x)

            # Calcular pérdida con sparsity_coef actualizado
            total_loss, mse_loss, sparsity_loss = loss_function(
                x, reconstruction, hidden, current_sparsity_coef
            )

            # Métricas
```

```
            l0 = (hidden > 0).float().sum(dim=1).mean().item()
            fraction_active = (hidden > 0).float().mean().item()

            # Acumular
            val_total_loss += total_loss.item()
            val_mse_loss += mse_loss.item()
            val_sparsity_loss += sparsity_loss.item()
            val_l0 += l0
            val_fraction_active += fraction_active

    # Promedios de validación
    num_val_batches = len(val_loader)
    avg_val_total_loss = val_total_loss / num_val_batches
    avg_val_mse_loss = val_mse_loss / num_val_batches
    avg_val_sparsity_loss = val_sparsity_loss / num_val_batches
    avg_val_l0 = val_l0 / num_val_batches
    avg_val_fraction_active = val_fraction_active / num_val_batches

    # Guardar en historial
    history['train_total_loss'].append(avg_train_total_loss)
    history['train_mse_loss'].append(avg_train_mse_loss)
    history['train_sparsity_loss'].append(avg_train_sparsity_loss)
    history['train_l0_sparsity'].append(avg_train_l0)
    history['train_fraction_active'].append(avg_train_fraction_active)

    history['val_total_loss'].append(avg_val_total_loss)
    history['val_mse_loss'].append(avg_val_mse_loss)
    history['val_sparsity_loss'].append(avg_val_sparsity_loss)
    history['val_l0_sparsity'].append(avg_val_l0)
    history['val_fraction_active'].append(avg_val_fraction_active)

    # ========== EARLY STOPPING ==========
    if CONFIG['early_stopping']:
        if avg_val_mse_loss < best_val_mse:
            best_val_mse = avg_val_mse_loss
            best_epoch = epoch + 1
            epochs_no_improve = 0

            # Guardar mejor modelo (local + organizado)
            best_checkpoint = {
                'model_state_dict': model.state_dict(),
                'optimizer_state_dict': optimizer.state_dict(),
```

```
            'config': CONFIG,
            'history': history,
            'epoch': best_epoch,
            'val_mse': best_val_mse
        }

        # Guardado local
        best_model_name = get_model_name(
            variante=NAMING_META['variante'],
            tecnica=NAMING_META['tecnica'],
            capa=NAMING_META['capa'],
            juegos=NAMING_META['juegos'],
            estado='best'
        )
        torch.save(best_checkpoint, CONFIG['save_dir'] / best_model_name)

        # Guardado organizado
        save_checkpoint(
            model,
            base_path=NAMING_META['base_path'],
            variante=NAMING_META['variante'],
            tecnica=NAMING_META['tecnica'],
            capa=NAMING_META['capa'],
            juegos=NAMING_META['juegos'],
            estado='best',
            extras=NAMING_META['extras'],
            config=CONFIG,
            history=history,
            optimizer=optimizer
        )

        print(f' Mejor modelo guardado en época {best_epoch} (Val MSE: {best_val_mse:.4f
    else:
        epochs_no_improve += 1
        if epochs_no_improve >= CONFIG['patience']:
            print(f'\n  Early stopping activado en época {epoch+1}')
            print(f'   Mejor modelo: época {best_epoch} con Val MSE = {best_val_mse:.4f}
            break

# Log cada 5 épocas
if (epoch + 1) % 5 == 0 or epoch == 0:
    print(f'\n Época {epoch+1}/{CONFIG["num_epochs"]} | L1 coef: {current_sparsity_coef:
```

```
        print(f'  Train - Total: {avg_train_total_loss:.4f} | MSE: {avg_train_mse_loss:.4f}
        print(f'  Val   - Total: {avg_val_total_loss:.4f} | MSE: {avg_val_mse_loss:.4f} | L0
        if CONFIG['early_stopping']:
            print(f'   Best Val MSE: {best_val_mse:.4f} @ época {best_epoch} | Sin mejora: {

print('\n' + '='*70)
print(' RESUMEN DEL ENTRENAMIENTO')
print('='*70)

if CONFIG['early_stopping']:
    print(f'Mejor modelo en época {best_epoch} con Val MSE = {best_val_mse:.4f}')
    print(f'Early stopping tras {len(history["train_total_loss"])} épocas')
else:
    print(f'Entrenamiento completo: {CONFIG["num_epochs"]} épocas')

# Tabla de métricas finales
final_metrics = pd.DataFrame({
    'Métrica': ['Total Loss', 'MSE Loss', 'L0 Sparsity', 'Fraction Active'],
    'Train': [
        f"{history['train_total_loss'][-1]:.4f}",
        f"{history['train_mse_loss'][-1]:.4f}",
        f"{history['train_l0_sparsity'][-1]:.1f}",
        f"{history['train_fraction_active'][-1]:.2%}"
    ],
    'Validation': [
        f"{history['val_total_loss'][-1]:.4f}",
        f"{history['val_mse_loss'][-1]:.4f}",
        f"{history['val_l0_sparsity'][-1]:.1f}",
        f"{history['val_fraction_active'][-1]:.2%}"
    ]
})

print('\nMétricas Finales (Última Época):')
print(final_metrics.to_string(index=False))
print('='*70)
```

```
 Iniciando entrenamiento: 200 épocas máximo
 L1 decay: 5.00e-04 × 0.95^época
  Early stopping: patience=15 épocas
======================================================================
```

```
Época 1/200 [Train] L1=5.00e-04: 100%|     | 47/47 [00:01<00:00, 46.08it/s, loss=0.4550, mse

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 1 (Val MSE: 0.3168)

 Época 1/200 | L1 coef: 5.00e-04
  Train - Total: 0.6495 | MSE: 0.5376 | L0: 1289.4
  Val   - Total: 0.4289 | MSE: 0.3168 | L0: 936.6
   Best Val MSE: 0.3168 @ época 1 | Sin mejora: 0/15

Época 2/200 [Train] L1=4.75e-04: 100%|     | 47/47 [00:00<00:00, 67.08it/s, loss=0.2577, mse

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 2 (Val MSE: 0.1603)

Época 3/200 [Train] L1=4.51e-04: 100%|     | 47/47 [00:00<00:00, 68.93it/s, loss=0.1981, mse

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 3 (Val MSE: 0.0990)

Época 4/200 [Train] L1=4.29e-04: 100%|     | 47/47 [00:00<00:00, 66.05it/s, loss=0.1627, mse

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 4 (Val MSE: 0.0732)

Época 5/200 [Train] L1=4.07e-04: 100%|     | 47/47 [00:00<00:00, 64.91it/s, loss=0.1429, mse

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 5 (Val MSE: 0.0617)

 Época 5/200 | L1 coef: 4.07e-04
  Train - Total: 0.1423 | MSE: 0.0563 | L0: 801.6
  Val   - Total: 0.1458 | MSE: 0.0617 | L0: 820.8
   Best Val MSE: 0.0617 @ época 5 | Sin mejora: 0/15
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Época 6/200 [Train] L1=3.87e-04: 100%|     | 47/47 [00:00<00:00, 66.57it/s, loss=0.1244, mse

  Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 6 (Val MSE: 0.0572)

Época 7/200 [Train] L1=3.68e-04: 100%|     | 47/47 [00:00<00:00, 66.66it/s, loss=0.1104, mse

  Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 7 (Val MSE: 0.0532)

Época 8/200 [Train] L1=3.49e-04: 100%|     | 47/47 [00:00<00:00, 68.52it/s, loss=0.1018, mse

  Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 8 (Val MSE: 0.0505)

Época 9/200 [Train] L1=3.32e-04: 100%|     | 47/47 [00:00<00:00, 65.10it/s, loss=0.0949, mse

  Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 9 (Val MSE: 0.0493)

Época 10/200 [Train] L1=3.15e-04: 100%|     | 47/47 [00:00<00:00, 62.51it/s, loss=0.0913, ms

  Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 10 (Val MSE: 0.0470)

 Época 10/200 | L1 coef: 3.15e-04
  Train - Total: 0.0910 | MSE: 0.0346 | L0: 809.1
  Val   - Total: 0.1031 | MSE: 0.0470 | L0: 843.9
   Best Val MSE: 0.0470 @ época 10 | Sin mejora: 0/15

Época 11/200 [Train] L1=2.99e-04: 100%|     | 47/47 [00:00<00:00, 67.03it/s, loss=0.0835, ms

  Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 11 (Val MSE: 0.0450)
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Época 12/200 [Train] L1=2.84e-04: 100%|     | 47/47 [00:00<00:00, 65.72it/s, loss=0.0753, ms

  Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 12 (Val MSE: 0.0439)

Época 13/200 [Train] L1=2.70e-04: 100%|     | 47/47 [00:00<00:00, 67.52it/s, loss=0.0765, ms

  Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 13 (Val MSE: 0.0429)

Época 14/200 [Train] L1=2.57e-04: 100%|     | 47/47 [00:00<00:00, 66.76it/s, loss=0.0717, ms

  Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 14 (Val MSE: 0.0413)

Época 15/200 [Train] L1=2.44e-04: 100%|     | 47/47 [00:00<00:00, 64.94it/s, loss=0.0691, ms

  Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 15 (Val MSE: 0.0406)

 Época 15/200 | L1 coef: 2.44e-04
  Train - Total: 0.0671 | MSE: 0.0260 | L0: 858.2
  Val   - Total: 0.0816 | MSE: 0.0406 | L0: 893.5
   Best Val MSE: 0.0406 @ época 15 | Sin mejora: 0/15

Época 16/200 [Train] L1=2.32e-04: 100%|     | 47/47 [00:00<00:00, 66.35it/s, loss=0.0614, ms

  Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 16 (Val MSE: 0.0399)

Época 17/200 [Train] L1=2.20e-04: 100%|     | 47/47 [00:00<00:00, 65.05it/s, loss=0.0600, ms

  Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 17 (Val MSE: 0.0387)
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Época 18/200 [Train] L1=2.09e-04: 100%|     | 47/47 [00:00<00:00, 65.14it/s, loss=0.0573, ms

  Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 18 (Val MSE: 0.0366)

Época 19/200 [Train] L1=1.99e-04: 100%|     | 47/47 [00:00<00:00, 66.97it/s, loss=0.0543, ms

  Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 19 (Val MSE: 0.0361)

Época 20/200 [Train] L1=1.89e-04: 100%|     | 47/47 [00:00<00:00, 67.20it/s, loss=0.0504, ms

  Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 20 (Val MSE: 0.0357)

 Época 20/200 | L1 coef: 1.89e-04
  Train - Total: 0.0515 | MSE: 0.0205 | L0: 899.5
  Val   - Total: 0.0670 | MSE: 0.0357 | L0: 936.7
   Best Val MSE: 0.0357 @ época 20 | Sin mejora: 0/15

Época 21/200 [Train] L1=1.79e-04: 100%|     | 47/47 [00:00<00:00, 64.61it/s, loss=0.0501, ms

  Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 21 (Val MSE: 0.0346)

Época 22/200 [Train] L1=1.70e-04: 100%|     | 47/47 [00:00<00:00, 67.12it/s, loss=0.0457, ms

  Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 22 (Val MSE: 0.0336)

Época 23/200 [Train] L1=1.62e-04: 100%|     | 47/47 [00:00<00:00, 67.80it/s, loss=0.0433, ms

  Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 23 (Val MSE: 0.0327)
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Época 24/200 [Train] L1=1.54e-04: 100%|     | 47/47 [00:00<00:00, 64.92it/s, loss=0.0431, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 24 (Val MSE: 0.0318)

Época 25/200 [Train] L1=1.46e-04: 100%|     | 47/47 [00:00<00:00, 68.54it/s, loss=0.0398, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 25 (Val MSE: 0.0311)

 Época 25/200 | L1 coef: 1.46e-04
  Train - Total: 0.0406 | MSE: 0.0165 | L0: 931.4
  Val   - Total: 0.0553 | MSE: 0.0311 | L0: 966.1
   Best Val MSE: 0.0311 @ época 25 | Sin mejora: 0/15

Época 26/200 [Train] L1=1.39e-04: 100%|     | 47/47 [00:00<00:00, 67.94it/s, loss=0.0387, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 26 (Val MSE: 0.0300)

Época 27/200 [Train] L1=1.32e-04: 100%|     | 47/47 [00:00<00:00, 64.93it/s, loss=0.0383, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 27 (Val MSE: 0.0296)

Época 28/200 [Train] L1=1.25e-04: 100%|     | 47/47 [00:00<00:00, 67.26it/s, loss=0.0377, ms
Época 29/200 [Train] L1=1.19e-04: 100%|     | 47/47 [00:00<00:00, 67.79it/s, loss=0.0356, ms
Época 30/200 [Train] L1=1.13e-04: 100%|     | 47/47 [00:00<00:00, 67.65it/s, loss=0.0326, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 30 (Val MSE: 0.0279)

 Época 30/200 | L1 coef: 1.13e-04
  Train - Total: 0.0341 | MSE: 0.0150 | L0: 947.6
  Val   - Total: 0.0472 | MSE: 0.0279 | L0: 981.2
   Best Val MSE: 0.0279 @ época 30 | Sin mejora: 0/15
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Época 31/200 [Train] L1=1.07e-04: 100%|     | 47/47 [00:00<00:00, 63.38it/s, loss=0.0298, ms

  Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 31 (Val MSE: 0.0267)

Época 32/200 [Train] L1=1.02e-04: 100%|     | 47/47 [00:00<00:00, 66.34it/s, loss=0.0297, ms

  Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 32 (Val MSE: 0.0261)

Época 33/200 [Train] L1=9.69e-05: 100%|     | 47/47 [00:00<00:00, 66.86it/s, loss=0.0285, ms

  Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 33 (Val MSE: 0.0257)

Época 34/200 [Train] L1=9.20e-05: 100%|     | 47/47 [00:00<00:00, 66.49it/s, loss=0.0269, ms

  Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 34 (Val MSE: 0.0249)

Época 35/200 [Train] L1=8.74e-05: 100%|     | 47/47 [00:00<00:00, 66.07it/s, loss=0.0250, ms

  Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 35 (Val MSE: 0.0243)

 Época 35/200 | L1 coef: 8.74e-05
  Train - Total: 0.0258 | MSE: 0.0107 | L0: 977.3
  Val   - Total: 0.0396 | MSE: 0.0243 | L0: 1010.0
   Best Val MSE: 0.0243 @ época 35 | Sin mejora: 0/15

Época 36/200 [Train] L1=8.30e-05: 100%|     | 47/47 [00:00<00:00, 67.48it/s, loss=0.0243, ms

  Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 36 (Val MSE: 0.0239)
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Época 37/200 [Train] L1=7.89e-05: 100%|     | 47/47 [00:00<00:00, 64.52it/s, loss=0.0236, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 37 (Val MSE: 0.0234)

Época 38/200 [Train] L1=7.49e-05: 100%|     | 47/47 [00:00<00:00, 66.48it/s, loss=0.0223, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 38 (Val MSE: 0.0232)

Época 39/200 [Train] L1=7.12e-05: 100%|     | 47/47 [00:00<00:00, 66.43it/s, loss=0.0218, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 39 (Val MSE: 0.0228)

Época 40/200 [Train] L1=6.76e-05: 100%|     | 47/47 [00:00<00:00, 65.85it/s, loss=0.0207, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 40 (Val MSE: 0.0224)

 Época 40/200 | L1 coef: 6.76e-05
  Train - Total: 0.0216 | MSE: 0.0096 | L0: 992.1
  Val   - Total: 0.0346 | MSE: 0.0224 | L0: 1022.8
   Best Val MSE: 0.0224 @ época 40 | Sin mejora: 0/15

Época 41/200 [Train] L1=6.43e-05: 100%|     | 47/47 [00:00<00:00, 67.31it/s, loss=0.0202, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 41 (Val MSE: 0.0220)

Época 42/200 [Train] L1=6.10e-05: 100%|     | 47/47 [00:00<00:00, 68.20it/s, loss=0.0204, ms
Época 43/200 [Train] L1=5.80e-05: 100%|     | 47/47 [00:00<00:00, 67.28it/s, loss=0.0189, ms
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 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 43 (Val MSE: 0.0210)

Época 44/200 [Train] L1=5.51e-05: 100%|     | 47/47 [00:00<00:00, 67.23it/s, loss=0.0167, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 44 (Val MSE: 0.0205)

Época 45/200 [Train] L1=5.23e-05: 100%|     | 47/47 [00:00<00:00, 66.56it/s, loss=0.0171, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 45 (Val MSE: 0.0200)

 Época 45/200 | L1 coef: 5.23e-05
  Train - Total: 0.0176 | MSE: 0.0079 | L0: 1001.3
  Val   - Total: 0.0298 | MSE: 0.0200 | L0: 1028.7
   Best Val MSE: 0.0200 @ época 45 | Sin mejora: 0/15

Época 46/200 [Train] L1=4.97e-05: 100%|     | 47/47 [00:00<00:00, 66.99it/s, loss=0.0162, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 46 (Val MSE: 0.0194)

Época 47/200 [Train] L1=4.72e-05: 100%|     | 47/47 [00:00<00:00, 66.39it/s, loss=0.0157, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 47 (Val MSE: 0.0190)

Época 48/200 [Train] L1=4.49e-05: 100%|     | 47/47 [00:00<00:00, 68.29it/s, loss=0.0148, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 48 (Val MSE: 0.0188)
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Época 49/200 [Train] L1=4.26e-05: 100%|     | 47/47 [00:00<00:00, 65.49it/s, loss=0.0145, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 49 (Val MSE: 0.0180)

Época 50/200 [Train] L1=4.05e-05: 100%|     | 47/47 [00:00<00:00, 67.28it/s, loss=0.0133, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 50 (Val MSE: 0.0177)

 Época 50/200 | L1 coef: 4.05e-05
  Train - Total: 0.0140 | MSE: 0.0063 | L0: 1008.2
  Val   - Total: 0.0255 | MSE: 0.0177 | L0: 1032.9
   Best Val MSE: 0.0177 @ época 50 | Sin mejora: 0/15

Época 51/200 [Train] L1=3.85e-05: 100%|     | 47/47 [00:00<00:00, 68.05it/s, loss=0.0130, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 51 (Val MSE: 0.0177)

Época 52/200 [Train] L1=3.65e-05: 100%|     | 47/47 [00:00<00:00, 66.14it/s, loss=0.0129, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 52 (Val MSE: 0.0171)

Época 53/200 [Train] L1=3.47e-05: 100%|     | 47/47 [00:00<00:00, 66.29it/s, loss=0.0123, ms
Época 54/200 [Train] L1=3.30e-05: 100%|     | 47/47 [00:00<00:00, 67.06it/s, loss=0.0121, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 54 (Val MSE: 0.0166)

Época 55/200 [Train] L1=3.13e-05: 100%|     | 47/47 [00:00<00:00, 67.25it/s, loss=0.0112, ms
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 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 55 (Val MSE: 0.0159)

 Época 55/200 | L1 coef: 3.13e-05
  Train - Total: 0.0121 | MSE: 0.0058 | L0: 1006.7
  Val   - Total: 0.0222 | MSE: 0.0159 | L0: 1029.9
   Best Val MSE: 0.0159 @ época 55 | Sin mejora: 0/15

Época 56/200 [Train] L1=2.98e-05: 100%|     | 47/47 [00:00<00:00, 66.02it/s, loss=0.0111, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 56 (Val MSE: 0.0156)

Época 57/200 [Train] L1=2.83e-05: 100%|     | 47/47 [00:00<00:00, 65.54it/s, loss=0.0101, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 57 (Val MSE: 0.0151)

Época 58/200 [Train] L1=2.69e-05: 100%|     | 47/47 [00:00<00:00, 65.78it/s, loss=0.0105, ms
Época 59/200 [Train] L1=2.55e-05: 100%|     | 47/47 [00:00<00:00, 63.70it/s, loss=0.0092, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 59 (Val MSE: 0.0144)

Época 60/200 [Train] L1=2.42e-05: 100%|     | 47/47 [00:00<00:00, 65.45it/s, loss=0.0098, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 60 (Val MSE: 0.0139)

 Época 60/200 | L1 coef: 2.42e-05
  Train - Total: 0.0098 | MSE: 0.0047 | L0: 1006.8
  Val   - Total: 0.0190 | MSE: 0.0139 | L0: 1026.2
   Best Val MSE: 0.0139 @ época 60 | Sin mejora: 0/15

Época 61/200 [Train] L1=2.30e-05: 100%|     | 47/47 [00:00<00:00, 66.36it/s, loss=0.0087, ms
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 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 61 (Val MSE: 0.0137)

Época 62/200 [Train] L1=2.19e-05: 100%|     | 47/47 [00:00<00:00, 64.38it/s, loss=0.0092, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 62 (Val MSE: 0.0131)

Época 63/200 [Train] L1=2.08e-05: 100%|     | 47/47 [00:00<00:00, 65.66it/s, loss=0.0081, ms
Época 64/200 [Train] L1=1.97e-05: 100%|     | 47/47 [00:00<00:00, 67.37it/s, loss=0.0081, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 64 (Val MSE: 0.0127)

Época 65/200 [Train] L1=1.88e-05: 100%|     | 47/47 [00:00<00:00, 64.53it/s, loss=0.0077, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 65 (Val MSE: 0.0121)

 Época 65/200 | L1 coef: 1.88e-05
  Train - Total: 0.0081 | MSE: 0.0040 | L0: 1001.4
  Val   - Total: 0.0162 | MSE: 0.0121 | L0: 1017.4
   Best Val MSE: 0.0121 @ época 65 | Sin mejora: 0/15

Época 66/200 [Train] L1=1.78e-05: 100%|     | 47/47 [00:00<00:00, 59.60it/s, loss=0.0089, ms
Época 67/200 [Train] L1=1.69e-05: 100%|     | 47/47 [00:00<00:00, 65.95it/s, loss=0.0078, ms
Época 68/200 [Train] L1=1.61e-05: 100%|     | 47/47 [00:00<00:00, 65.60it/s, loss=0.0073, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 68 (Val MSE: 0.0114)

Época 69/200 [Train] L1=1.53e-05: 100%|     | 47/47 [00:00<00:00, 64.74it/s, loss=0.0067, ms
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 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 69 (Val MSE: 0.0111)

Época 70/200 [Train] L1=1.45e-05: 100%|     | 47/47 [00:00<00:00, 66.71it/s, loss=0.0065, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 70 (Val MSE: 0.0106)

 Época 70/200 | L1 coef: 1.45e-05
  Train - Total: 0.0069 | MSE: 0.0036 | L0: 988.0
  Val   - Total: 0.0140 | MSE: 0.0106 | L0: 1001.2
   Best Val MSE: 0.0106 @ época 70 | Sin mejora: 0/15

Época 71/200 [Train] L1=1.38e-05: 100%|     | 47/47 [00:00<00:00, 63.65it/s, loss=0.0069, ms
Época 72/200 [Train] L1=1.31e-05: 100%|     | 47/47 [00:00<00:00, 65.48it/s, loss=0.0062, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 72 (Val MSE: 0.0105)

Época 73/200 [Train] L1=1.24e-05: 100%|     | 47/47 [00:00<00:00, 67.02it/s, loss=0.0059, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 73 (Val MSE: 0.0100)

Época 74/200 [Train] L1=1.18e-05: 100%|     | 47/47 [00:00<00:00, 66.75it/s, loss=0.0055, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 74 (Val MSE: 0.0098)

Época 75/200 [Train] L1=1.12e-05: 100%|     | 47/47 [00:00<00:00, 64.42it/s, loss=0.0052, ms
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 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 75 (Val MSE: 0.0091)

 Época 75/200 | L1 coef: 1.12e-05
  Train - Total: 0.0056 | MSE: 0.0028 | L0: 980.5
  Val   - Total: 0.0118 | MSE: 0.0091 | L0: 989.2
   Best Val MSE: 0.0091 @ época 75 | Sin mejora: 0/15

Época 76/200 [Train] L1=1.07e-05: 100%|     | 47/47 [00:00<00:00, 64.52it/s, loss=0.0047, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 76 (Val MSE: 0.0087)

Época 77/200 [Train] L1=1.01e-05: 100%|     | 47/47 [00:00<00:00, 57.60it/s, loss=0.0054, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 77 (Val MSE: 0.0085)

Época 78/200 [Train] L1=9.63e-06: 100%|     | 47/47 [00:00<00:00, 59.10it/s, loss=0.0047, ms
Época 79/200 [Train] L1=9.15e-06: 100%|     | 47/47 [00:00<00:00, 58.33it/s, loss=0.0045, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 79 (Val MSE: 0.0085)

Época 80/200 [Train] L1=8.69e-06: 100%|     | 47/47 [00:00<00:00, 56.20it/s, loss=0.0043, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 80 (Val MSE: 0.0083)

 Época 80/200 | L1 coef: 8.69e-06
  Train - Total: 0.0047 | MSE: 0.0025 | L0: 975.1
  Val   - Total: 0.0105 | MSE: 0.0083 | L0: 982.7
   Best Val MSE: 0.0083 @ época 80 | Sin mejora: 0/15

Época 81/200 [Train] L1=8.26e-06: 100%|     | 47/47 [00:00<00:00, 57.55it/s, loss=0.0057, ms
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 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 81 (Val MSE: 0.0079)

Época 82/200 [Train] L1=7.84e-06: 100%|     | 47/47 [00:00<00:00, 57.95it/s, loss=0.0048, ms
Época 83/200 [Train] L1=7.45e-06: 100%|     | 47/47 [00:00<00:00, 60.25it/s, loss=0.0040, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 83 (Val MSE: 0.0074)

Época 84/200 [Train] L1=7.08e-06: 100%|     | 47/47 [00:00<00:00, 59.85it/s, loss=0.0043, ms
Época 85/200 [Train] L1=6.73e-06: 100%|     | 47/47 [00:00<00:00, 58.69it/s, loss=0.0040, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 85 (Val MSE: 0.0070)

 Época 85/200 | L1 coef: 6.73e-06
  Train - Total: 0.0042 | MSE: 0.0024 | L0: 968.1
  Val   - Total: 0.0088 | MSE: 0.0070 | L0: 974.1
   Best Val MSE: 0.0070 @ época 85 | Sin mejora: 0/15

Época 86/200 [Train] L1=6.39e-06: 100%|     | 47/47 [00:00<00:00, 59.50it/s, loss=0.0036, ms
Época 87/200 [Train] L1=6.07e-06: 100%|     | 47/47 [00:00<00:00, 59.23it/s, loss=0.0038, ms
Época 88/200 [Train] L1=5.77e-06: 100%|     | 47/47 [00:00<00:00, 57.70it/s, loss=0.0033, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 88 (Val MSE: 0.0067)

Época 89/200 [Train] L1=5.48e-06: 100%|     | 47/47 [00:00<00:00, 58.31it/s, loss=0.0036, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 89 (Val MSE: 0.0063)

Época 90/200 [Train] L1=5.20e-06: 100%|     | 47/47 [00:00<00:00, 59.13it/s, loss=0.0035, ms
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 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 90 (Val MSE: 0.0063)

 Época 90/200 | L1 coef: 5.20e-06
  Train - Total: 0.0037 | MSE: 0.0022 | L0: 964.9
  Val   - Total: 0.0077 | MSE: 0.0063 | L0: 968.6
   Best Val MSE: 0.0063 @ época 90 | Sin mejora: 0/15

Época 91/200 [Train] L1=4.94e-06: 100%|     | 47/47 [00:00<00:00, 59.97it/s, loss=0.0030, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 91 (Val MSE: 0.0061)

Época 92/200 [Train] L1=4.70e-06: 100%|     | 47/47 [00:00<00:00, 56.50it/s, loss=0.0030, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 92 (Val MSE: 0.0057)

Época 93/200 [Train] L1=4.46e-06: 100%|     | 47/47 [00:00<00:00, 56.68it/s, loss=0.0029, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 93 (Val MSE: 0.0056)

Época 94/200 [Train] L1=4.24e-06: 100%|     | 47/47 [00:00<00:00, 58.58it/s, loss=0.0027, ms

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 94 (Val MSE: 0.0055)

Época 95/200 [Train] L1=4.03e-06: 100%|     | 47/47 [00:00<00:00, 56.29it/s, loss=0.0023, ms
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 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 95 (Val MSE: 0.0050)

 Época 95/200 | L1 coef: 4.03e-06
  Train - Total: 0.0029 | MSE: 0.0017 | L0: 963.2
  Val   - Total: 0.0062 | MSE: 0.0050 | L0: 967.9
   Best Val MSE: 0.0050 @ época 95 | Sin mejora: 0/15

Época 96/200 [Train] L1=3.83e-06: 100%|     | 47/47 [00:00<00:00, 56.61it/s, loss=0.0023, ms
Época 97/200 [Train] L1=3.63e-06: 100%|     | 47/47 [00:00<00:00, 57.72it/s, loss=0.0027, ms
Época 98/200 [Train] L1=3.45e-06: 100%|     | 47/47 [00:00<00:00, 57.72it/s, loss=0.0024, ms
Época 99/200 [Train] L1=3.28e-06: 100%|     | 47/47 [00:00<00:00, 55.37it/s, loss=0.0027, ms
Época 100/200 [Train] L1=3.12e-06: 100%|     | 47/47 [00:00<00:00, 59.00it/s, loss=0.0024, m


 Época 100/200 | L1 coef: 3.12e-06
  Train - Total: 0.0028 | MSE: 0.0019 | L0: 960.1
  Val   - Total: 0.0060 | MSE: 0.0051 | L0: 963.8
   Best Val MSE: 0.0050 @ época 95 | Sin mejora: 5/15

Época 101/200 [Train] L1=2.96e-06: 100%|     | 47/47 [00:00<00:00, 59.04it/s, loss=0.0023, m
Época 102/200 [Train] L1=2.81e-06: 100%|     | 47/47 [00:00<00:00, 56.51it/s, loss=0.0022, m

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 102 (Val MSE: 0.0050)

Época 103/200 [Train] L1=2.67e-06: 100%|     | 47/47 [00:00<00:00, 57.23it/s, loss=0.0023, m
Época 104/200 [Train] L1=2.54e-06: 100%|     | 47/47 [00:00<00:00, 55.93it/s, loss=0.0020, m

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 104 (Val MSE: 0.0046)

Época 105/200 [Train] L1=2.41e-06: 100%|     | 47/47 [00:00<00:00, 57.52it/s, loss=0.0018, m
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 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 105 (Val MSE: 0.0043)

 Época 105/200 | L1 coef: 2.41e-06
  Train - Total: 0.0024 | MSE: 0.0017 | L0: 959.0
  Val   - Total: 0.0051 | MSE: 0.0043 | L0: 962.4
   Best Val MSE: 0.0043 @ época 105 | Sin mejora: 0/15


Época 106/200 [Train] L1=2.29e-06: 100%|     | 47/47 [00:00<00:00, 59.61it/s, loss=0.0019, m

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 106 (Val MSE: 0.0041)


Época 107/200 [Train] L1=2.18e-06: 100%|     | 47/47 [00:00<00:00, 61.09it/s, loss=0.0019, m
Época 108/200 [Train] L1=2.07e-06: 100%|     | 47/47 [00:00<00:00, 58.49it/s, loss=0.0018, m
Época 109/200 [Train] L1=1.96e-06: 100%|     | 47/47 [00:00<00:00, 58.63it/s, loss=0.0018, m
Época 110/200 [Train] L1=1.87e-06: 100%|     | 47/47 [00:00<00:00, 52.80it/s, loss=0.0016, m

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 110 (Val MSE: 0.0039)

 Época 110/200 | L1 coef: 1.87e-06
  Train - Total: 0.0022 | MSE: 0.0016 | L0: 961.4
  Val   - Total: 0.0045 | MSE: 0.0039 | L0: 962.5
   Best Val MSE: 0.0039 @ época 110 | Sin mejora: 0/15


Época 111/200 [Train] L1=1.77e-06: 100%|     | 47/47 [00:00<00:00, 59.67it/s, loss=0.0020, m
Época 112/200 [Train] L1=1.68e-06: 100%|     | 47/47 [00:00<00:00, 61.55it/s, loss=0.0017, m
Época 113/200 [Train] L1=1.60e-06: 100%|     | 47/47 [00:00<00:00, 65.82it/s, loss=0.0018, m

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 113 (Val MSE: 0.0037)


Época 114/200 [Train] L1=1.52e-06: 100%|     | 47/47 [00:00<00:00, 66.28it/s, loss=0.0017, m
Época 115/200 [Train] L1=1.44e-06: 100%|     | 47/47 [00:00<00:00, 65.99it/s, loss=0.0022, m
```

```
 Época 115/200 | L1 coef: 1.44e-06
  Train - Total: 0.0021 | MSE: 0.0017 | L0: 960.7
  Val   - Total: 0.0042 | MSE: 0.0038 | L0: 961.6
   Best Val MSE: 0.0037 @ época 113 | Sin mejora: 2/15

Época 116/200 [Train] L1=1.37e-06: 100%|     | 47/47 [00:00<00:00, 66.46it/s, loss=0.0018, m
Época 117/200 [Train] L1=1.30e-06: 100%|     | 47/47 [00:00<00:00, 63.42it/s, loss=0.0017, m

  Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 117 (Val MSE: 0.0037)

Época 118/200 [Train] L1=1.24e-06: 100%|     | 47/47 [00:00<00:00, 65.97it/s, loss=0.0016, m

  Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 118 (Val MSE: 0.0036)

Época 119/200 [Train] L1=1.18e-06: 100%|     | 47/47 [00:00<00:00, 66.04it/s, loss=0.0016, m

  Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 119 (Val MSE: 0.0034)

Época 120/200 [Train] L1=1.12e-06: 100%|     | 47/47 [00:00<00:00, 62.84it/s, loss=0.0018, m


 Época 120/200 | L1 coef: 1.12e-06
  Train - Total: 0.0020 | MSE: 0.0016 | L0: 960.4
  Val   - Total: 0.0041 | MSE: 0.0037 | L0: 960.7
   Best Val MSE: 0.0034 @ época 119 | Sin mejora: 1/15

Época 121/200 [Train] L1=1.06e-06: 100%|     | 47/47 [00:00<00:00, 66.14it/s, loss=0.0013, m

  Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 121 (Val MSE: 0.0031)
```

```
Época 122/200 [Train] L1=1.01e-06: 100%|     | 47/47 [00:00<00:00, 66.50it/s, loss=0.0018, m
Época 123/200 [Train] L1=9.58e-07: 100%|     | 47/47 [00:00<00:00, 64.24it/s, loss=0.0014, m
Época 124/200 [Train] L1=9.10e-07: 100%|     | 47/47 [00:00<00:00, 64.77it/s, loss=0.0014, m
Época 125/200 [Train] L1=8.64e-07: 100%|     | 47/47 [00:00<00:00, 65.45it/s, loss=0.0013, m


 Época 125/200 | L1 coef: 8.64e-07
  Train - Total: 0.0018 | MSE: 0.0015 | L0: 961.6
  Val   - Total: 0.0035 | MSE: 0.0032 | L0: 960.7
   Best Val MSE: 0.0031 @ época 121 | Sin mejora: 4/15

Época 126/200 [Train] L1=8.21e-07: 100%|     | 47/47 [00:00<00:00, 63.68it/s, loss=0.0011, m

  Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 126 (Val MSE: 0.0030)

Época 127/200 [Train] L1=7.80e-07: 100%|     | 47/47 [00:00<00:00, 57.11it/s, loss=0.0013, m

  Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 127 (Val MSE: 0.0030)

Época 128/200 [Train] L1=7.41e-07: 100%|     | 47/47 [00:00<00:00, 59.17it/s, loss=0.0014, m
Época 129/200 [Train] L1=7.04e-07: 100%|     | 47/47 [00:00<00:00, 58.86it/s, loss=0.0010, m

  Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 129 (Val MSE: 0.0028)

Época 130/200 [Train] L1=6.69e-07: 100%|     | 47/47 [00:00<00:00, 58.79it/s, loss=0.0013, m


 Época 130/200 | L1 coef: 6.69e-07
  Train - Total: 0.0014 | MSE: 0.0012 | L0: 964.2
  Val   - Total: 0.0034 | MSE: 0.0031 | L0: 960.7
   Best Val MSE: 0.0028 @ época 129 | Sin mejora: 1/15

Época 131/200 [Train] L1=6.35e-07: 100%|     | 47/47 [00:00<00:00, 58.95it/s, loss=0.0011, m
```

```
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 131 (Val MSE: 0.0028)

Época 132/200 [Train] L1=6.04e-07: 100%|     | 47/47 [00:00<00:00, 54.12it/s, loss=0.0013, m
Época 133/200 [Train] L1=5.73e-07: 100%|     | 47/47 [00:00<00:00, 57.98it/s, loss=0.0012, m
Época 134/200 [Train] L1=5.45e-07: 100%|     | 47/47 [00:00<00:00, 57.46it/s, loss=0.0014, m

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 134 (Val MSE: 0.0027)

Época 135/200 [Train] L1=5.18e-07: 100%|     | 47/47 [00:00<00:00, 57.74it/s, loss=0.0016, m


 Época 135/200 | L1 coef: 5.18e-07
  Train - Total: 0.0016 | MSE: 0.0014 | L0: 964.0
  Val   - Total: 0.0030 | MSE: 0.0028 | L0: 965.3
   Best Val MSE: 0.0027 @ época 134 | Sin mejora: 1/15

Época 136/200 [Train] L1=4.92e-07: 100%|     | 47/47 [00:00<00:00, 55.81it/s, loss=0.0017, m
Época 137/200 [Train] L1=4.67e-07: 100%|     | 47/47 [00:00<00:00, 58.51it/s, loss=0.0014, m
Época 138/200 [Train] L1=4.44e-07: 100%|     | 47/47 [00:00<00:00, 57.96it/s, loss=0.0014, m
Época 139/200 [Train] L1=4.22e-07: 100%|     | 47/47 [00:00<00:00, 56.79it/s, loss=0.0013, m

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 139 (Val MSE: 0.0026)

Época 140/200 [Train] L1=4.00e-07: 100%|     | 47/47 [00:00<00:00, 62.30it/s, loss=0.0010, m

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 140 (Val MSE: 0.0023)

 Época 140/200 | L1 coef: 4.00e-07
  Train - Total: 0.0013 | MSE: 0.0012 | L0: 965.8
  Val   - Total: 0.0024 | MSE: 0.0023 | L0: 964.9
   Best Val MSE: 0.0023 @ época 140 | Sin mejora: 0/15
```

```
Época 141/200 [Train] L1=3.80e-07: 100%|     | 47/47 [00:00<00:00, 60.99it/s, loss=0.0011, m
Época 142/200 [Train] L1=3.61e-07: 100%|     | 47/47 [00:00<00:00, 61.94it/s, loss=0.0009, m
Época 143/200 [Train] L1=3.43e-07: 100%|     | 47/47 [00:00<00:00, 50.72it/s, loss=0.0008, m

  Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 143 (Val MSE: 0.0021)

Época 144/200 [Train] L1=3.26e-07: 100%|     | 47/47 [00:00<00:00, 58.15it/s, loss=0.0008, m

  Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 144 (Val MSE: 0.0021)

Época 145/200 [Train] L1=3.10e-07: 100%|     | 47/47 [00:00<00:00, 57.69it/s, loss=0.0009, m


 Época 145/200 | L1 coef: 3.10e-07
  Train - Total: 0.0011 | MSE: 0.0010 | L0: 968.0
  Val   - Total: 0.0023 | MSE: 0.0022 | L0: 965.7
   Best Val MSE: 0.0021 @ época 144 | Sin mejora: 1/15

Época 146/200 [Train] L1=2.94e-07: 100%|     | 47/47 [00:00<00:00, 59.20it/s, loss=0.0009, m
Época 147/200 [Train] L1=2.80e-07: 100%|     | 47/47 [00:00<00:00, 59.59it/s, loss=0.0012, m
Época 148/200 [Train] L1=2.66e-07: 100%|     | 47/47 [00:00<00:00, 57.93it/s, loss=0.0008, m

  Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 148 (Val MSE: 0.0021)

Época 149/200 [Train] L1=2.52e-07: 100%|     | 47/47 [00:00<00:00, 56.78it/s, loss=0.0014, m
Época 150/200 [Train] L1=2.40e-07: 100%|     | 47/47 [00:00<00:00, 58.17it/s, loss=0.0012, m


 Época 150/200 | L1 coef: 2.40e-07
  Train - Total: 0.0016 | MSE: 0.0015 | L0: 966.4
  Val   - Total: 0.0025 | MSE: 0.0024 | L0: 964.7
   Best Val MSE: 0.0021 @ época 148 | Sin mejora: 2/15
```

```
Época 151/200 [Train] L1=2.28e-07: 100%|     | 47/47 [00:00<00:00, 58.18it/s, loss=0.0008, m
Época 152/200 [Train] L1=2.16e-07: 100%|     | 47/47 [00:00<00:00, 59.26it/s, loss=0.0009, m
Época 153/200 [Train] L1=2.06e-07: 100%|     | 47/47 [00:00<00:00, 57.36it/s, loss=0.0007, m
Época 154/200 [Train] L1=1.95e-07: 100%|     | 47/47 [00:00<00:00, 56.83it/s, loss=0.0008, m

  Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 154 (Val MSE: 0.0020)

Época 155/200 [Train] L1=1.86e-07: 100%|     | 47/47 [00:00<00:00, 60.47it/s, loss=0.0012, m


 Época 155/200 | L1 coef: 1.86e-07
  Train - Total: 0.0011 | MSE: 0.0011 | L0: 967.6
  Val   - Total: 0.0026 | MSE: 0.0025 | L0: 966.0
   Best Val MSE: 0.0020 @ época 154 | Sin mejora: 1/15

Época 156/200 [Train] L1=1.76e-07: 100%|     | 47/47 [00:00<00:00, 54.19it/s, loss=0.0009, m
Época 157/200 [Train] L1=1.67e-07: 100%|     | 47/47 [00:00<00:00, 65.49it/s, loss=0.0007, m
Época 158/200 [Train] L1=1.59e-07: 100%|     | 47/47 [00:00<00:00, 65.17it/s, loss=0.0008, m
Época 159/200 [Train] L1=1.51e-07: 100%|     | 47/47 [00:00<00:00, 66.38it/s, loss=0.0011, m
Época 160/200 [Train] L1=1.44e-07: 100%|     | 47/47 [00:00<00:00, 63.51it/s, loss=0.0008, m


 Época 160/200 | L1 coef: 1.44e-07
  Train - Total: 0.0011 | MSE: 0.0011 | L0: 967.7
  Val   - Total: 0.0022 | MSE: 0.0022 | L0: 966.2
   Best Val MSE: 0.0020 @ época 154 | Sin mejora: 6/15

Época 161/200 [Train] L1=1.36e-07: 100%|     | 47/47 [00:00<00:00, 64.03it/s, loss=0.0012, m
Época 162/200 [Train] L1=1.30e-07: 100%|     | 47/47 [00:00<00:00, 58.99it/s, loss=0.0008, m

  Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 162 (Val MSE: 0.0019)

Época 163/200 [Train] L1=1.23e-07: 100%|     | 47/47 [00:00<00:00, 53.13it/s, loss=0.0009, m
Época 164/200 [Train] L1=1.17e-07: 100%|     | 47/47 [00:00<00:00, 55.15it/s, loss=0.0008, m
```

```
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 164 (Val MSE: 0.0019)

Época 165/200 [Train] L1=1.11e-07: 100%|     | 47/47 [00:00<00:00, 56.76it/s, loss=0.0014, m


 Época 165/200 | L1 coef: 1.11e-07
  Train - Total: 0.0011 | MSE: 0.0011 | L0: 968.8
  Val   - Total: 0.0035 | MSE: 0.0035 | L0: 965.9
   Best Val MSE: 0.0019 @ época 164 | Sin mejora: 1/15

Época 166/200 [Train] L1=1.06e-07: 100%|     | 47/47 [00:00<00:00, 61.12it/s, loss=0.0010, m
Época 167/200 [Train] L1=1.00e-07: 100%|     | 47/47 [00:00<00:00, 56.30it/s, loss=0.0007, m

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 167 (Val MSE: 0.0018)

Época 168/200 [Train] L1=9.52e-08: 100%|     | 47/47 [00:00<00:00, 56.27it/s, loss=0.0006, m

 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 168 (Val MSE: 0.0017)

Época 169/200 [Train] L1=9.05e-08: 100%|     | 47/47 [00:00<00:00, 60.07it/s, loss=0.0006, m
Época 170/200 [Train] L1=8.60e-08: 100%|     | 47/47 [00:00<00:00, 59.36it/s, loss=0.0010, m


 Época 170/200 | L1 coef: 8.60e-08
  Train - Total: 0.0010 | MSE: 0.0010 | L0: 969.5
  Val   - Total: 0.0021 | MSE: 0.0021 | L0: 966.5
   Best Val MSE: 0.0017 @ época 168 | Sin mejora: 2/15

Época 171/200 [Train] L1=8.17e-08: 100%|     | 47/47 [00:00<00:00, 58.45it/s, loss=0.0007, m
Época 172/200 [Train] L1=7.76e-08: 100%|     | 47/47 [00:00<00:00, 58.58it/s, loss=0.0007, m
Época 173/200 [Train] L1=7.37e-08: 100%|     | 47/47 [00:00<00:00, 60.05it/s, loss=0.0008, m
Época 174/200 [Train] L1=7.00e-08: 100%|     | 47/47 [00:00<00:00, 59.50it/s, loss=0.0007, m
Época 175/200 [Train] L1=6.65e-08: 100%|     | 47/47 [00:00<00:00, 58.35it/s, loss=0.0008, m
```

```
 Época 175/200 | L1 coef: 6.65e-08
  Train - Total: 0.0008 | MSE: 0.0008 | L0: 970.4
  Val   - Total: 0.0020 | MSE: 0.0019 | L0: 968.2
   Best Val MSE: 0.0017 @ época 168 | Sin mejora: 7/15

Época 176/200 [Train] L1=6.32e-08: 100%|     | 47/47 [00:00<00:00, 60.52it/s, loss=0.0012, m
Época 177/200 [Train] L1=6.00e-08: 100%|     | 47/47 [00:00<00:00, 57.96it/s, loss=0.0009, m
Época 178/200 [Train] L1=5.70e-08: 100%|     | 47/47 [00:00<00:00, 56.72it/s, loss=0.0007, m
Época 179/200 [Train] L1=5.42e-08: 100%|     | 47/47 [00:00<00:00, 57.50it/s, loss=0.0005, m

  Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_best.pt
 Mejor modelo guardado en época 179 (Val MSE: 0.0015)

Época 180/200 [Train] L1=5.15e-08: 100%|     | 47/47 [00:00<00:00, 59.26it/s, loss=0.0006, m


 Época 180/200 | L1 coef: 5.15e-08
  Train - Total: 0.0010 | MSE: 0.0009 | L0: 969.9
  Val   - Total: 0.0018 | MSE: 0.0018 | L0: 966.2
   Best Val MSE: 0.0015 @ época 179 | Sin mejora: 1/15

Época 181/200 [Train] L1=4.89e-08: 100%|     | 47/47 [00:00<00:00, 58.58it/s, loss=0.0008, m
Época 182/200 [Train] L1=4.64e-08: 100%|     | 47/47 [00:00<00:00, 64.96it/s, loss=0.0007, m
Época 183/200 [Train] L1=4.41e-08: 100%|     | 47/47 [00:00<00:00, 65.70it/s, loss=0.0010, m
Época 184/200 [Train] L1=4.19e-08: 100%|     | 47/47 [00:00<00:00, 66.59it/s, loss=0.0010, m
Época 185/200 [Train] L1=3.98e-08: 100%|     | 47/47 [00:00<00:00, 65.92it/s, loss=0.0007, m


 Época 185/200 | L1 coef: 3.98e-08
  Train - Total: 0.0013 | MSE: 0.0012 | L0: 967.1
  Val   - Total: 0.0017 | MSE: 0.0017 | L0: 966.2
   Best Val MSE: 0.0015 @ época 179 | Sin mejora: 6/15

Época 186/200 [Train] L1=3.78e-08: 100%|     | 47/47 [00:00<00:00, 58.91it/s, loss=0.0008, m
Época 187/200 [Train] L1=3.59e-08: 100%|     | 47/47 [00:00<00:00, 63.72it/s, loss=0.0006, m
Época 188/200 [Train] L1=3.41e-08: 100%|     | 47/47 [00:00<00:00, 66.69it/s, loss=0.0007, m
Época 189/200 [Train] L1=3.24e-08: 100%|     | 47/47 [00:00<00:00, 66.64it/s, loss=0.0010, m
Época 190/200 [Train] L1=3.08e-08: 100%|     | 47/47 [00:00<00:00, 65.03it/s, loss=0.0007, m
```

```
 Época 190/200 | L1 coef: 3.08e-08
  Train - Total: 0.0013 | MSE: 0.0013 | L0: 968.0
  Val   - Total: 0.0018 | MSE: 0.0018 | L0: 965.9
   Best Val MSE: 0.0015 @ época 179 | Sin mejora: 11/15

Época 191/200 [Train] L1=2.93e-08: 100%|     | 47/47 [00:00<00:00, 66.50it/s, loss=0.0008, m
Época 192/200 [Train] L1=2.78e-08: 100%|     | 47/47 [00:00<00:00, 66.72it/s, loss=0.0012, m
Época 193/200 [Train] L1=2.64e-08: 100%|     | 47/47 [00:00<00:00, 66.57it/s, loss=0.0006, m
Época 194/200 [Train] L1=2.51e-08: 100%|     | 47/47 [00:00<00:00, 63.56it/s, loss=0.0009, m


   Early stopping activado en época 194
   Mejor modelo: época 179 con Val MSE = 0.0015

======================================================================
 RESUMEN DEL ENTRENAMIENTO
======================================================================
Mejor modelo en época 179 con Val MSE = 0.0015
Early stopping tras 194 épocas

Métricas Finales (Última Época):
        Métrica  Train Validation
     Total Loss 0.0008     0.0018
       MSE Loss 0.0008     0.0018
    L0 Sparsity  970.4      967.2
Fraction Active  5.92%      5.90%
======================================================================
```

8. Guardar Modelo

```python
# Guardar checkpoint final (local + organizado)
checkpoint = {
    'model_state_dict': model.state_dict(),
    'optimizer_state_dict': optimizer.state_dict(),
    'config': CONFIG,
    'history': history
}

# Guardado local
```

```python
final_model_name = get_model_name(
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],
    capa=NAMING_META['capa'],
    juegos=NAMING_META['juegos'],
    estado='final'
)
checkpoint_path = CONFIG['save_dir'] / final_model_name
torch.save(checkpoint, checkpoint_path)

# Guardado organizado
save_checkpoint(
    model,
    base_path=NAMING_META['base_path'],
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],
    capa=NAMING_META['capa'],
    juegos=NAMING_META['juegos'],
    estado='final',
    extras=NAMING_META['extras'],
    config=CONFIG,
    history=history,
    optimizer=optimizer
)

# Resumen de archivos guardados
saved_files = pd.DataFrame({
    'Tipo': ['Local', 'Organizado'],
    'Ubicación': [
        str(checkpoint_path),
        f"layer_{NAMING_META['capa']:02d}/models/..."
    ]
})

print('\n' + '='*70)
print(' CHECKPOINTS GUARDADOS')
print('='*70)
print(saved_files.to_string(index=False))
print('='*70)
```

```
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex5_final.pt
```

```
==================================================================
 CHECKPOINTS GUARDADOS
==================================================================
      Tipo                                              Ubicación
     Local saved_model\sae_standard_l1-decay_l6_500g_final.pt
Organizado                                   layer_06/models/...
==================================================================
```

9. Métricas básicas del modelo

```
# Guardar metricas de entrenamiento
metrics_dir = get_metrics_dir(
    NAMING_META["experiment_base_path"],
    NAMING_META["variante"],
    NAMING_META["tecnica"],
    NAMING_META["capa"],
    NAMING_META["juegos"],
    extras_id=extras_id
)

training_metrics = {
    # Metricas de calidad
    "val_mse_final": history["val_mse_loss"][-1],
    "val_mse_best": best_val_mse,
    "val_l0_final": history["val_l0_sparsity"][-1],
    "val_fraction_active_final": history["val_fraction_active"][-1],
    # Contexto del entrenamiento
    "best_epoch": best_epoch,
    "total_epochs": len(history["train_total_loss"]),
    "train_mse_final": history["train_mse_loss"][-1],
    "train_l0_final": history["train_l0_sparsity"][-1],
    # Hiperparametros
    "hidden_dim": CONFIG["hidden_dim"],
    "expansion_factor": CONFIG["hidden_dim"] / CONFIG["input_dim"],
    "sparsity_coef": CONFIG["sparsity_coef"],
    "sparsity_decay": CONFIG["sparsity_decay"],
    "learning_rate": CONFIG["learning_rate"],
    "num_games": NAMING_META["juegos"],
}
```

```
metrics_path = metrics_dir / "training_metrics.json"
with open(metrics_path, "w") as f:
    _json.dump(training_metrics, f, indent=2)

# Resumen
metrics_df = pd.DataFrame({
    "Metrica": ["Val MSE (final)", "Val MSE (mejor)", "Val L0 (final)", "Fraccion Activa"],
    "Valor": [
        f"{training_metrics['val_mse_final']:.6f}",
        f"{training_metrics['val_mse_best']:.6f}",
        f"{training_metrics['val_l0_final']:.1f}",
        f"{training_metrics['val_fraction_active_final']:.2%}"
    ]
})
print("" + "="*50)
print(" METRICAS DE ENTRENAMIENTO GUARDADAS")
print("="*50)
print(metrics_df.to_string(index=False))
print(f"Archivo: {metrics_path}")
print("="*50)
```

```
==================================================
 METRICAS DE ENTRENAMIENTO GUARDADAS
==================================================
        Metrica    Valor
Val MSE (final) 0.001790
Val MSE (mejor) 0.001509
 Val L0 (final)    967.2
Fraccion Activa    5.90%
Archivo: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard\
decay_standard\500games\ex5\metrics\training_metrics.json
==================================================
```

10. Visualización de Resultados

```
# Gráficas de pérdidas (Train vs Validation) + L1 Decay
fig, axes = plt.subplots(2, 3, figsize=(18, 10))

# Total loss
axes[0, 0].plot(history['train_total_loss'], label='Train', alpha=0.8)
```

```
axes[0, 0].plot(history['val_total_loss'], label='Validation', alpha=0.8)
axes[0, 0].set_title('Total Loss')
axes[0, 0].set_xlabel('Época')
axes[0, 0].set_ylabel('Loss')
axes[0, 0].legend()
axes[0, 0].grid(True, alpha=0.3)

# MSE loss
axes[0, 1].plot(history['train_mse_loss'], label='Train', color='orange', alpha=0.8)
axes[0, 1].plot(history['val_mse_loss'], label='Validation', color='red', alpha=0.8)
if CONFIG['early_stopping'] and best_epoch > 0:
    axes[0, 1].axvline(best_epoch - 1, color='green', linestyle='--', alpha=0.7, label=f'Bes
axes[0, 1].set_title('MSE Loss (Reconstrucción)')
axes[0, 1].set_xlabel('Época')
axes[0, 1].set_ylabel('MSE')
axes[0, 1].legend()
axes[0, 1].grid(True, alpha=0.3)

# L0 sparsity
axes[0, 2].plot(history['train_l0_sparsity'], label='Train', color='green', alpha=0.8)
axes[0, 2].plot(history['val_l0_sparsity'], label='Validation', color='darkgreen', alpha=0.8
axes[0, 2].set_title('L0 Sparsity (Features Activas)')
axes[0, 2].set_xlabel('Época')
axes[0, 2].set_ylabel('Número de features')
axes[0, 2].legend()
axes[0, 2].grid(True, alpha=0.3)

# Fraction active
axes[1, 0].plot(history['train_fraction_active'], label='Train', color='purple', alpha=0.8)
axes[1, 0].plot(history['val_fraction_active'], label='Validation', color='magenta', alpha=0
axes[1, 0].set_title('Fracción de Features Activas')
axes[1, 0].set_xlabel('Época')
axes[1, 0].set_ylabel('Fracción')
axes[1, 0].legend()
axes[1, 0].grid(True, alpha=0.3)

# Sparsity Coefficient (L1 Decay)
axes[1, 1].plot(history['sparsity_coef_history'], color='brown', linewidth=2)
axes[1, 1].set_title('L1 Coefficient (Decay)')
axes[1, 1].set_xlabel('Época')
axes[1, 1].set_ylabel('Sparsity Coef')
axes[1, 1].set_yscale('log')
```

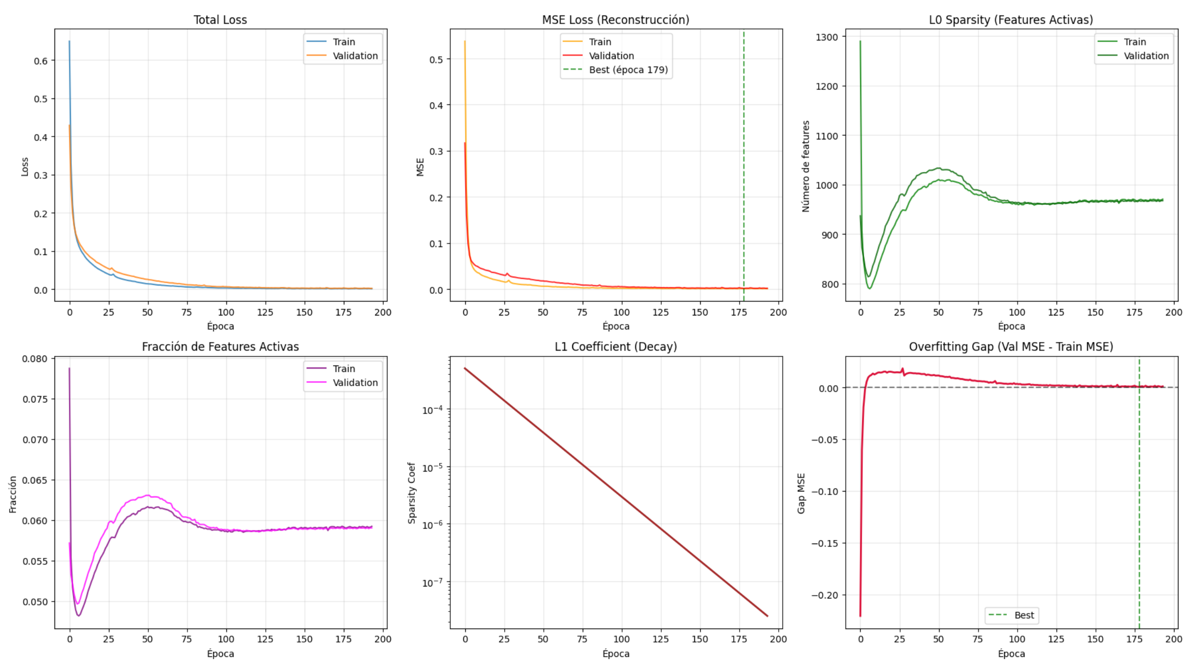
```
axes[1, 1].grid(True, alpha=0.3)

# Train vs Val MSE Gap
mse_gap = [v - t for v, t in zip(history['val_mse_loss'], history['train_mse_loss'])]
axes[1, 2].plot(mse_gap, color='crimson', linewidth=2)
axes[1, 2].axhline(0, color='black', linestyle='--', alpha=0.5)
if CONFIG['early_stopping'] and best_epoch > 0:
    axes[1, 2].axvline(best_epoch - 1, color='green', linestyle='--', alpha=0.7, label=f'Bes
axes[1, 2].set_title('Overfitting Gap (Val MSE - Train MSE)')
axes[1, 2].set_xlabel('Época')
axes[1, 2].set_ylabel('Gap MSE')
axes[1, 2].legend()
axes[1, 2].grid(True, alpha=0.3)

plt.tight_layout()
plt.savefig(CONFIG['plots_dir'] / 'training_curves.png', dpi=300, bbox_inches='tight')
plt.show()

print('\n' + '='*50)
print('   Gráficas guardadas exitosamente')

print('='*50)
```



```
==================================================
  Gráficas guardadas exitosamente
==================================================
```

11. Resumen Final

```
# Resumen final completo
print('\n' + '='*70)
print(' RESUMEN FINAL DEL ENTRENAMIENTO')
print('='*70)

# Arquitectura
arch_summary = pd.DataFrame({
    'Componente': ['Input', 'Hidden', 'Parámetros'],
    'Valor': [
        f"{CONFIG['input_dim']}",
        f"{CONFIG['hidden_dim']} ({CONFIG['hidden_dim']/CONFIG['input_dim']:.1f}x)",
        f"{num_params:,}"
    ]
})
print('\nArquitectura:')
print(arch_summary.to_string(index=False))

# Datos
data_summary = pd.DataFrame({
    'Conjunto': ['Train', 'Validation', 'Total'],
    'Muestras': [
        f"{len(train_dataset):,}",
        f"{len(val_dataset):,}",
        f"{len(full_dataset):,}"
    ]
})
print('\nDatos:')
print(data_summary.to_string(index=False))

# Métricas finales
final_summary = pd.DataFrame({
    'Métrica': ['Train Loss', 'Val Loss', 'Train MSE', 'Val MSE', 'Train L0', 'Val L0'],
    'Valor': [
        f"{history['train_total_loss'][-1]:.4f}",
```

```
        f"{history['val_total_loss'][-1]:.4f}",
        f"{history['train_mse_loss'][-1]:.4f}",
        f"{history['val_mse_loss'][-1]:.4f}",
        f"{history['train_l0_sparsity'][-1]:.1f}",
        f"{history['val_l0_sparsity'][-1]:.1f}"
    ]
})
print('\nMétricas Finales:')
print(final_summary.to_string(index=False))

# Archivos
files_summary = pd.DataFrame({
    'Tipo': ['Modelo', 'Gráficas'],
    'Ubicación': [
        str(checkpoint_path),
        str(CONFIG['plots_dir'] / 'training_curves.png')
    ]
})
print('\nArchivos Guardados:')
print(files_summary.to_string(index=False))

print('\n' + '='*70)
print('   ENTRENAMIENTO COMPLETADO EXITOSAMENTE')
print('='*70)
```

```
======================================================================
 RESUMEN FINAL DEL ENTRENAMIENTO
======================================================================

Arquitectura:
Componente         Valor
     Input           512
    Hidden 16384 (32.0x)
Parámetros    16,794,112

Datos:
  Conjunto Muestras
     Train   23,600
Validation    5,900
     Total   29,500
```

```
Métricas Finales:
   Métrica  Valor
Train Loss 0.0008
  Val Loss 0.0018
 Train MSE 0.0008
   Val MSE 0.0018
  Train L0  970.4
    Val L0  967.2

Archivos Guardados:
    Tipo
  Modelo
decay_l6_500g_final.pt
Gráficas c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard\
decay_standard\500games\ex5\plots\training_curves.png

======================================================================
  ENTRENAMIENTO COMPLETADO EXITOSAMENTE
======================================================================
```

12. Generar PDF con Quarto

Una vez ejecutado todo el notebook, ejecutar el siguiente comando en la terminal para generar el PDF con Quarto:

```python
import subprocess

# Obtener directorio del experimento y nombre del PDF
exp_dir = get_experiment_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)

pdf_name = get_report_name(
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
```

```
    'training',
    extras_id=extras_id
)

notebook_name = 'train_sae_standard_l1-decay.ipynb'
output_path = exp_dir / pdf_name

print(f' Generando PDF con Quarto...')
print(f' Archivo de salida: {output_path}')

# Quarto no acepta rutas en --output, solo nombre de archivo
# Se usa --output-dir para el directorio y --output solo para el nombre
result = subprocess.run(
    f'quarto render {notebook_name} --to pdf --output-dir "{exp_dir}" --output {pdf_name}',
    shell=True,
    capture_output=True,
    text=True
)

if result.returncode == 0:
    print(f' PDF generado exitosamente: {output_path}')
else:
    print(f' Error al generar PDF:')
    print(result.stderr)
```

```
 Generando PDF con Quarto...
 Archivo de salida: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sa
decay_standard\500games\ex5\sae_standard_l1-decay_l6_500g_ex5_training.pdf
 PDF generado exitosamente: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\lay
decay_standard\500games\ex5\sae_standard_l1-decay_l6_500g_ex5_training.pdf
```